Use Case Specification

Green Points Platform × Tasco VETC

Social & Mobility Track | Hackathon 2025

Document Information

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Sponsor Track	Social & Mobility by TASCO
Features covered	UC-01 Live Green Ring | UC-02 CO_2 Meter | UC-03 Tasco Dashboard | UC-04 Smart Route

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UC-01 Live Green Ring — Daily Green Gamification

UC-01 Live Green Ring — Daily Green Gamification	
Use Case ID	UC-01
Use Case Name	Live Green Ring — Daily Green Gamification
Feature	Live Green Ring
Actor(s)	Registered User (Anh Minh / Chị Lan persona), Green Points System, VETC API
Priority	Must Have — P0 (Core MVP)
Trigger	
Preconditions	1. User has registered and linked VETC account 2. User has set home → workplace route (A → B) during onboarding 3. Vehicle type is configured (motorbike / car / EV) 4. Location permission granted (background) 5. Push notification permission granted
Postconditions	1. Daily ring reflects accurate completion % based on verified green actions 2. Green Points balance is updated in real-time after each task 3. Streak counter incremented if ring reaches ≥ 70% by 11:59 PM 4. CO_2 contribution data logged to Tasco Dashboard (UC-03) 5. If ring = 100%: celebration animation shown, bonus 10 pts awarded
Main flow(Basic path)	1. System generates 3 personalized tasks at 6:00 AM based on user's A→B route *→ Tasks generated: (a) Use ETC through toll booth, (b) Depart before 7:30 AM (off-peak), (c) Park within first 5 minutes* 2. User opens app → sees ring at 0% with today's 3 task cards 3. User commutes; VETC API detects toll transaction → webhook fires within 3 seconds 4. System validates ETC event: match user ID + timestamp + expected toll gate on route 5. Task (a) auto-completes → ring fills +33% → push notification sent: 'ETC qua trạm — tiết kiệm 0.06 kg CO_2. +5 Green Points!' 6. User departs before 7:30 AM (GPS timestamp) → task (b) auto-completes → ring +33% 7. Parking event detected via payment API → task (c) validated → ring +34% → ring FULL 8. Full ring celebration: confetti animation, 'Hôm nay bạn đã đóng góp 0.18 kg CO_2 — tương đương trồng 0.018 cây xanh!' 9. Green Points (15 pts base + 10 pts streak bonus) credited to wallet 10. Data pushed to Tasco B2B Dashboard aggregate (anonymous)
Alternative flow 1	ALT 1A — User does not use ETC (pays cash at toll): 3a. VETC API receives no transaction for expected toll gate 4a. After 15-minute timeout, task (a) remains incomplete 5a. Push notification: 'Dùng VETC lần sau để nhận Green Points và giảm CO_2!' 6a. Ring can still reach 100% via remaining 2 tasks ALT 1B — User takes different route (no toll booth): 3b. GPS detects route deviation from A→B plan 4b. System substitutes task (a) with: 'Drive ≤ 50 km/h average speed (eco-driving)'

	5b. Speed data tracked via phone GPS → task validated at destination arrival
Alternative flow 2	ALT 2 — Ring incomplete by 11:59 PM (< 70%): 1. No streak increment; streak counter resets to 0 2. System logs partial contribution (proportional CO_2 credit still applied) 3. Push notification next morning: 'Hôm qua bạn hoàn thành 2/3 tasks. Hôm nay thử hoàn thành cả 3 nhé!'
Exception flows	EX-01: VETC API timeout (> 5 sec) → retry 3 times → if fail, task marked 'pending verification' → resolved within 1 hour via batch sync EX-02: GPS signal lost mid-route → fallback to last known position + estimated arrival; task completion held until GPS restored EX-03: User has no route set → ring shows 'generic' tasks (use ETC anywhere, drive off-peak) instead of route-specific tasks
Business rules	BR-01: 1 ring per calendar day; resets at midnight local time BR-02: Tasks are route-specific; if no commute day (weekend), ring switches to 'Leisure mode' with different tasks BR-03: Green Points expiry: 12 months from issuance BR-04: Minimum 1 task completed for any CO_2 data to be logged BR-05: Anti-gaming: same VETC transaction cannot complete tasks for 2 different users
Related use cases	UC-02 (CO_2 Meter feeds task validation), UC-03 (Tasco Dashboard receives aggregate data), UC-04 (Smart Route may adjust task difficulty)

UC-02 Real-Time CO_2 Meter — Live Emission Display While Driving

UC-02 Real-Time CO_2 Meter — Live Emission Display While Driving	
Use Case ID	UC-02
Use Case Name	Real-Time CO_2 Meter — Live Emission Display While Driving
Feature	Real-Time CO_2 Meter
Actor(s)	Driver (User), GPS Module, Vehicle Profile, Carbon Calculation Engine
Priority	Must Have — P0 (Core differentiator)
Trigger	User taps 'Start Journey' OR app auto-detects motion > 20 km/h for 2 minutes
Preconditions	1. User is authenticated and vehicle type is set 2. Location (foreground) and motion permissions granted 3. Journey destination entered OR route A→B loaded from profile 4. Device GPS signal strength ≥ 2 bars (accuracy ≤ 50m) 5. App is in foreground (or background with location permission)
Postconditions	1. Trip CO_2 log saved: distance (km), duration, CO_2 emitted (kg), CO_2 saved vs baseline 2. Live Green Ring task data updated if ETC event occurred during journey 3. Weekly/monthly cumulative CO_2 chart updated on Impact screen 4. Trip data contributed to Tasco Scope 3 aggregate (UC-03) 5. If EV vehicle: 'CO_2 vs petrol equivalent' saved for social sharing
Main flow(Basic path)	1. User taps 'Start Journey'; enters destination or selects saved A→B route 2. System loads vehicle emission factor: xe máy xăng = 0.07 kgCO_2/km | ô tô xăng = 0.21 | xe điện = 0.02 (EVN grid) 3. GPS begins tracking at 1-second intervals; speed and coordinates logged 4. Dashboard displays live meter: current speed, CO_2/km (real-time), trip total CO_2 so far → *Display format: '2.4 km → 0.17 kg CO_2 emitted | Eco score: B+'* 5. At each kilometer: micro-update animation; running total refreshed 6. ETC gate detected (GPS coordinates match known gate location): idle time = 0 saved → CO_2 saved delta calculated and displayed in green: '-0.06 kg CO_2 saved!' 7. User arrives at destination → Journey ends automatically (speed < 5 km/h for 60 sec) 8. Trip summary shown: distance, total CO_2, CO_2 saved vs no-ETC scenario, eco-grade (A–F), Green Points earned 9. Translation displayed: '0.42 kg CO_2 = năng lượng để đun sôi 5 ấm nước' 10. User taps 'Save & Share' or dismisses; data persisted to local + Tasco cloud
Alternative flow 1	ALT 1 — User is driving EV (xe điện): 2a. Emission factor = 0.02 kgCO_2/km (EVN grid 2024: 0.5 kgCO_2/kWh ÷ avg 25 km/kWh) 4a. Meter shows: 'CO_2 actual: 0.05 kg | vs petrol equivalent: 0.53 kg | You saved: 0.48 kg' 8a. Trip summary adds: 'Bạn đã tiết kiệm 0.48 kg CO_2 so với xe xăng tương đương. Tương đương 4.8 cây xanh tháng này!'
Alternative flow 2	ALT 2 — Traffic jam detected (avg speed < 15 km/h for > 5 min):

	3a. System flags 'congestion mode'; idle emission rate applied: +0.008 $kgCO_2$/min 4a. Meter turns amber: 'Đang kẹt xe — đang tiêu thụ nhiên liệu khi đứng yên' 5a. Suggestion popup: 'Tắt máy nếu dừng > 1 phút để tiết kiệm CO_2' (only shown once per trip)
Exception flows	EX-01: GPS signal lost > 30 sec → meter paused, 'Đang tìm tín hiệu GPS...' shown; resumes on signal restore; gap estimated via dead reckoning EX-02: App killed by OS (background restrictions) → trip resumed from last checkpoint on re-open; user prompted: 'Tiếp tục chuyến đi lúc 8:23?' EX-03: Speed > 130 km/h detected → data flagged as anomalous; emission calculation suspended; alert: 'Tốc độ bất thường — vui lòng kiểm tra'
Business rules	BR-01: Emission factors updated quarterly from MONRE / EPA MOVES database BR-02: Minimum trip distance = 0.5 km for data to be logged (filters GPS noise) BR-03: CO_2 saved = [(baseline_idle_time × 0.03 $kgCO_2$/min) + (ETC_passes × avg_idle_avoided_min × 0.03)] BR-04: Eco-grade scale: A (< 0.05 kg/km) | B (0.05–0.10) | C (0.10–0.15) | D (0.15–0.20) | F (> 0.20) BR-05: Data retention: raw GPS deleted after 30 days; aggregated trip metrics kept indefinitely
Related use cases	UC-01 (feeds task completion data to Live Green Ring), UC-03 (trip data contributes to Tasco Scope 3), UC-04 (Smart Route uses CO_2/km data for route scoring)

UC-03 Tasco B2B Dashboard — Corporate Carbon ESG Reporting

UC-03 Tasco B2B Dashboard — Corporate Carbon ESG Reporting	
Use Case ID	UC-03
Use Case Name	Tasco B2B Dashboard — Corporate Carbon ESG Reporting
Feature	Tasco B2B Dashboard
Actor(s)	Tasco ESG Officer (primary), Tasco Executive (viewer), Green Points Platform (data source), Auditor (read-only export consumer)
Priority	Must Have — P0 (Core B2B value)
Trigger	ESG Officer logs into dashboard.greenpoints.vn with corporate credentials
Preconditions	1. Tasco corporate account is provisioned with admin credentials 2. VETC transaction API integration is active and syncing 3. At least 1 active user has completed ≥ 1 green action in the reporting period 4. Emission factors database is loaded (EPA MOVES + EVN grid factor) 5. GHG Protocol reporting period is configured (default: calendar month)
Postconditions	1. Dashboard reflects real-time aggregate CO_2 reduction across all users 2. ESG Officer can export GHG Protocol Scope 3 report (PDF + CSV) in < 30 seconds 3. All exported reports are version-stamped with methodology reference 4. Dashboard data is audit-ready: each metric traceable to source transactions 5. Tasco internal ESG report preparation time reduced from ~3 days to < 1 hour
Main flow(Basic path)	1. ESG Officer navigates to dashboard.greenpoints.vn → authenticates via Tasco SSO 2. Overview panel loads: KPI cards showing: Total CO_2 reduced this month (kg), Active contributors, ETC transactions processed, vs. prior month delta 3. Officer selects reporting period (e.g., Q2 2025) from date picker 4. System renders aggregate charts: *→ Chart A: Daily CO_2 reduction trend (bar chart)* *→ Chart B: Breakdown by source — ETC idle-saving (68%), EV users (22%), off-peak driving (10%)* *→ Chart C: User vehicle type distribution (xe máy / ô tô xăng / xe điện)* 5. Officer drills into 'ETC contribution' tile → sees trip-level breakdown (anonymised: no PII) 6. Officer clicks 'Generate Report' → selects format: GHG Protocol Scope 3 / Internal ESG / CSV raw data 7. System generates PDF with: methodology statement, emission factors used, data period, total CO_2 reduction figure, limitation notes 8. PDF stamped: 'Data source: Green Points Platform v1.0 | Methodology: GHG Protocol Scope 3 Category 11 | Prepared: [date]' 9. Officer downloads and attaches to DJSI submission package 10. System logs export event (timestamp, user, format) for audit trail
Alternative flow 1	ALT 1 — External auditor access request: 1a. ESG Officer creates time-limited read-only link (expires in 7 days) 2a. Auditor accesses link → sees dashboard in 'Audit mode' (no export, no edits) 3a. Auditor verifies: sample ETC transactions visible with anonymised trip IDs

	4a. Auditor can request raw data export → ESG Officer approves → CSV generated
Alternative flow 2	ALT 2 — Data gap detected (< 80% expected ETC transactions synced): 1a. Banner shown: 'Dữ liệu VETC chưa đồng bộ đầy đủ — có thể ảnh hưởng đến độ chính xác báo cáo' 2a. System shows last successful sync timestamp 3a. ESG Officer can trigger manual sync or proceed with available data (report will note data completeness %)
Exception flows	EX-01: VETC API sync failure > 24 hours → alert sent to Tasco IT + ESG Officer; dashboard shows data age warning EX-02: Report generation timeout (large date range) → async generation; email notification when ready EX-03: Conflicting emission factor versions → system uses latest approved version; prior reports unaffected
Business rules	BR-01: All user data in dashboard is aggregated and anonymised — no individual PII visible to Tasco BR-02: CO_2 reduction methodology: GHG Protocol Scope 3, Category 11 (Use of sold products / services) BR-03: Baseline for ETC idle-saving: avg. 3.2 minutes idle per manual toll cash transaction × 0.03 $kgCO_2$/min BR-04: Dashboard data refresh rate: every 15 minutes (near real-time) BR-05: Exported reports must include 'Limitation: Individual carbon contributions are Green Points incentives, not certified carbon credits under VN ETS. Suitable for voluntary ESG disclosure.' BR-06: Data available for export: up to 3 years historical rolling window
Related use cases	UC-01 (data source: ring completions), UC-02 (data source: trip CO_2 logs), UC-04 (route efficiency data contributes to aggregate)

UC-04 Smart Route Suggestion — ETC-Optimized Green Routing

UC-04 Smart Route Suggestion — ETC-Optimized Green Routing	
Use Case ID	UC-04
Use Case Name	Smart Route Suggestion — ETC-Optimized Green Routing
Feature	Smart Route Suggestion for Drivers
Actor(s)	Driver (User), Routing Engine, Maps API (Google Maps / HERE), VETC Toll Data, Carbon Calculation Engine
Priority	Should Have — P1 (V1.1 post-MVP)
Trigger	User opens 'Plan Journey' screen OR opens app with pending A→B route and taps 'Suggest Route'
Preconditions	1. User authenticated with vehicle type and fuel type configured 2. Origin and destination entered (or loaded from saved profile) 3. Internet connection available for Maps API call 4. Maps API key active with Directions + Roads API enabled 5. VETC toll database loaded (gate locations, average idle time per gate)
Postconditions	1. User is presented with 2–3 route options, each with CO_2, time, cost, and Green Points scored 2. Selected route is saved to journey plan; UC-02 (CO_2 Meter) uses this route as baseline 3. If ETC-optimised route is selected: expected Green Points pre-credited as 'pending' (confirmed on completion) 4. Route deviation during trip: UC-04 recalculates and offers re-route suggestion
Main flow(Basic path)	1. User enters origin A and destination B → taps 'Find Green Routes' 2. System calls Maps API → retrieves 3 candidate routes (fastest / balanced / eco) 3. For each route, system calculates: *→ Distance (km) × vehicle emission factor = CO_2 emitted* *→ Number of ETC toll gates on route × idle_time_saved × 0.03 $kgCO_2$/min = CO_2 saved* *→ Estimated fuel cost (VND) based on distance + avg consumption* *→ Green Points earnable if ETC used at all gates on route* *→ Travel time (minutes) from Maps API* 4. Routes ranked by 'Green Score' = (CO_2_saved / CO_2_emitted) × 100 5. UI shows route comparison card: *→ Route 1 (Fastest): 22 min | 0.42 kg CO_2 | 0 ETC gates | 0 pts potential* *→ Route 2 (Balanced): 26 min | 0.38 kg CO_2 | 1 ETC gate | +8 pts potential ★ Recommended* *→ Route 3 (Greenest): 31 min | 0.31 kg CO_2 | 2 ETC gates | +15 pts potential* 6. User selects route → 'Start Journey' button activates UC-02 CO_2 Meter 7. During trip: if user deviates > 500m from selected route → gentle notification: 'Bạn đang đi khỏi tuyến đường xanh. Muốn quay lại?' 8. On arrival: actual vs estimated CO_2 compared; discrepancy > 20% flagged for model improvement
Alternative flow 1	ALT 1 — No ETC toll gates on any candidate route: 3a. Green Score calculated on distance and speed profile alone

	4a. Route recommendation based on lowest CO_2/km (favours less congested roads) 5a. UI note: 'Tuyến đường này không có trạm ETC — điểm xanh sẽ từ eco-driving'
Alternative flow 2	ALT 2 — User is driving EV (xe điện): 3a. Emission factor = 0.02 kgCO_2/km; ETC idle-saving carbon benefit de-weighted 4a. 'Greenest' route is almost always recommended (EV baseline so low that time/ETC matter less) 5a. UI shows: 'Xe điện của bạn đã tiết kiệm X kg CO_2 so với tuyến đường xăng tương đương'
Exception flows	EX-01: Maps API rate limit or timeout → show cached route from last 24h with age warning; disable live traffic estimate EX-02: Route calculation returns > 90 minutes → show simplified 2-option comparison + note 'Chuyến đi dài — cân nhắc dừng sạc/nghỉ' EX-03: Origin or destination outside VETC coverage zone → fallback to CO_2-only scoring (no ETC benefit shown)
Business rules	BR-01: Maximum 3 routes shown (UX constraint); always includes fastest option even if not greenest BR-02: 'Recommended' badge only assigned if Green Route saves ≥ 10% CO_2 vs fastest AND adds < 15 minutes BR-03: Green Points shown as 'potential' — only confirmed after UC-01 / UC-02 validates actual ETC usage BR-04: VETC gate locations sourced from Tasco VETC API (authoritative); updated weekly BR-05: Idle time per manual toll assumed = 3.2 minutes (from Tasco/NUS study baseline); configurable per gate BR-06: Route re-calculation triggered if: user deviates > 500m OR journey paused > 10 minutes
Related use cases	UC-01 (route tasks generated from selected route), UC-02 (CO_2 Meter uses route as baseline), UC-03 (route efficiency data feeds Tasco aggregate)